

Beyond the charger count: a spatial equity lens on local dynamics of public EV charging infrastructure rollout in London

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Abstract

Despite a substantial increase in the overall number of public EV charging points, the spatial disparity of deployments continues to hinder the supply of infrastructure to meet the varied demand. This calls for a deeper understanding of socio-spatial inequality and models counting supply-demand balance for future planning. Our analysis takes London as a case study and investigates the changes in accessibility to public charging points between 2021 and 2024. We found that despite supply growth outpacing demand, accessibility in some areas has paradoxically declined. We analysed five ward-level outcomes: public charging accessibility in 2021, in 2024, the accessibility changes between these years, EV charging supply variation, and EV adoption variation. Significant LM-Lag and LM-Error test results confirm spatial dependence, justifying Spatial Autoregressive (SAR) estimation to capture city-wide spillover effects. However, SAR's global coefficients mask important local heterogeneity. Geographically Weighted Regression (GWR) maps reveal that wealthier wards (higher median house prices) consistently experience faster charger deployment and increased EV uptake, while deprived or lower social-grade areas lag behind. These findings highlight that a purely market-driven rollout risks exacerbating transport disadvantage.

KEYWORDS: Transport; Electric vehicle; Electric vehicle charging stations; charging stations accessibility; Spatial justice.

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Abstract

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1. Introduction

The transition to electric vehicles (EVs) represents a cornerstone of the UK's climate strategy (Department for Transport, 2022, Tyers et al., 2024, Goddard, 2024), especially in cities, as thanks to their high population and employment density, economic hubs, and developed infrastructure, urban areas represent an ideal setting for promoting (and analysing) the transition to electric transportation (Kennedy et al., 2017, Kennedy et al., 2018, Romero-Lankao et al., 2021).

Under current policy, the sale of new petrol and diesel cars is to be banned by 2030, and all new cars must be zero-emission by 2035 (Geospatial Commission, 2022). Achieving these targets requires not only scaling up the public charging network - from roughly 35,000 chargepoints (in 2022 - the year of the Geospatial Commission's report publication) to at least 300,000 by 2030 - but also ensuring that chargers are allocated in ways that promote equitable access.

A supply-centric strategy risks reinforcing existing socio-spatial inequalities (Sovacool et al., 2019, Golub et al., 2019, Cui et al., 2019) if charge points primarily benefit the mobility needs of well-off households with access to private driveways (Caulfield et al., 2022, Romero-Lankao et al., 2022). Those without off-street parking - such as many council housing residents - and lower-income groups face higher barriers to charging, compounding transport disadvantage and limiting modal choice (Stacey and Shaddock, 2015, Carroll et al., 2021).

Transport disadvantage is strongly linked to income poverty and can exacerbate social and economic exclusion (Nie et al., 2024). Car dependence often becomes a self-reinforcing cycle, whereby households without access to reliable public transport or affordable mobility options must rely on costly private vehicles (Mattioli, 2014). For many lower-income households, the combined burdens of housing and transport costs already consume a disproportionate share of income (Lipman, 2006), and the shift to EVs threatens to increase this burden unless charging infrastructure is carefully planned. In addition, self-employed workers - such as tradespeople who use light vans - may face elevated costs when switching to electric models, given their high mileage and reliance on daytime charging (Sovacool and Dworkin, 2015). Without targeted interventions, the green transition could inadvertently deepen transport poverty for vulnerable groups (Carroll et al., 2021).

A growing literature examines the spatial distribution of chargepoints, focussing on aggregate supply metrics (Zhang et al., 2024b, Peng et al., 2024), demand modelling (Chen et al., 2015, Farkas et al., 2018, Qian et al., 2019, Liao et al., 2023, Pan et al., 2024), different types of accessibility (Hsu and Fingerman, 2021, Li et al., 2022, Gazmeh et al., 2024, Esmaili et al., 2024), or on technical siting optimisation (Baouche et al., 2014, Chen et al., 2020b, Gupta et al., 2021, Bao and Xie, 2021). For instance, focusing on London, Zhang et al. (2024b) analysed on-street household EV charging from an equity perspective, but primarily on supply clusters within and out of emission zones, leaving the demand side under-explored. Consequently, considering the London case study, important questions remain about how socio-economic, housing, and market factors drive both EV charging deployment (supply) and EV uptake (demand) at finer spatial scales.

Moreover, to move beyond aggregate measures of supply and demand, there is a need for a better understanding of the personal and place-based determinants that shape the transition to electric vehicle use. In particular, factors such as climate-change awareness and perceived environmental impact influence individual propensity to adopt EVs (Steg and Vlek, 2009, Graham-Rowe et al., 2012), while financial incentives and policy interventions - ranging from purchase subsidies to residential charging grants - have been shown to drive uptake in specific neighbourhoods (Breetz et al., 2018, Zhang et al., 2024a). Equally important are "who you are" (socio-demographic characteristics such as income, education, and household composition) and "where you live" (urban morphology, housing type and proximity to amenities), since these jointly determine access to, and suitability of, charging infrastructure.

This study addresses that gap by combining global and local spatial methods to evaluate where

inequities persist in London's charging network over two time points (2021 and 2024 – chosen on the basis of temporal trends in charging infrastructure developments and data availability). We first run a Spatial Autoregressive (SAR) analysis, which accounts for city-wide “spillover” effects, capturing how charger deployment or EV registrations in one area influence neighbouring areas. However, SAR yields a single coefficient for each predictor and cannot reveal local variation. To uncover such heterogeneity, we employ Geographically Weighted Regression (GWR), mapping local R^2 values and local coefficients for five dependent variables: (1) public EVCI accessibility in 2021; (2) public EVCI accessibility in 2024; (3) the change in accessibility between 2021 and 2024, (4) public EV charging infrastructure (EVCI) supply variation between 2021 and 2024; and (5) EV adoption (proxy for public EVCI demand) variation between 2021 and 2024. The GWR framework highlights where selected most critical predictors, ranging from car ownership and social grade to median house price, population density, and housing tenure, has varied associations with supply and demand outcomes across geographical areas.

London provides an ideal case study due to its socio-economic diversity, rapid EV rollout, and availability of fine-scale ward-level data on charger locations, vehicle registrations, and demographic indicators. We incorporate indicators of transport disadvantage - such as car ownership rates and deprivation indices - to test how equity concerns manifest across different boroughs. Building on the principles of energy justice, which call for fair distribution of both benefits and burdens in the energy transition (Lipman, 2006, Sovacool and Dworkin, 2015, Dawkins et al., 2023), we ask: do current strategies for charging infrastructure sufficiently address equity? And if not, where should policymakers focus interventions to ensure that socially and economically vulnerable communities are not left behind?

Our contributions are threefold. First, by comparing 2021 and 2024 accessibility, we identify neighbourhoods where equity gaps are closing or widening, thereby offering a dynamic view of infrastructure deployment relative to demand. Second, by running LM diagnostics and applying SAR models, we confirmed the presence of spatial effects in the scenario of EV transition. Third, our GWR analysis produced maps at fine spatial granularity, revealing the nuanced spatial patterns of both supply- and demand-side drivers within Greater London to support place-based decision making. The remainder of the paper is structured as follows: Section 2 describes the data sources and variables, and models specifications; Section 3 presents the OLS results, LM test results, SAR outputs, and GWR maps; Section 4 discusses the implications for transport equity, housing policy, and sustainable mobility, policy recommendations and future research direction; finally, Section 5 presents the conclusions of the study.

2. Data & methodology

Our analysis draws on a diverse array of spatialised datasets (Table 1) to characterise both the supply and demand for public EV charge-points and the socio-demographic context of London's Lower Layer Super Output Areas (LSOAs). Supply of public Electric Vehicle Charging Infrastructure (EVCI) is quantified as the count of publicly accessible charge-points per LSOA, sourced from the Department for Transport's National Charge Point Registry, while demand is proxied by the number of licensed EVs per LSOA, based on combined Department for Transport (DfT) and Driver and Vehicle Licensing Agency (DVLA) vehicle licensing records. To capture socio-economic variation, we employ the Approximated Social Grade (ASG) classification derived from the 2021 National Readership Survey and disseminated via the ONS, which segments residents into higher (A, B, C1) versus lower (C2, D, E) occupational grades. Deprivation is measured as the count of households experiencing deficits in three or more dimensions (education, employment, health, housing) from the 2021 Census, and car-ownership profiles distinguish between zero- or one-car households and those with two or more vehicles, also from Census data. Housing tenure (owned vs rented) and dwelling typology (flats, terraced, detached and semi-detached, and other) are likewise drawn from the Census, while local

market conditions are reflected in median house prices for 2021 and 2023 from the ONS House Price Statistics for Small Areas (HPSSA). Finally, population density (residents per hectare) is obtained from ONS demographic tables, and the concentration of amenities - captured as points-of-interest density - is calculated from OpenStreetMap extracts. Collectively, these layers provide a socio-spatial portrait for interrogating equity in EVCI accessibility across London.

Figure 1.d shows the growing trends of both EV licensing and deployment of public charging infrastructure in London between 2012 and 2024. While the number of licensed EVs has been steadily growing in the past fifteen years, its growth has been outpaced by the supply of public chargepoints in London after 2021. The scope of our analysis is to investigate whether such dramatic increase of supply is reflected in an actual growth of accessibility for everyone, with a particular focus on historically deprived communities, as they often face a “double jeopardy” in the electric mobility transition: lower household incomes and higher transport costs limit their ability to purchase EVs or install home chargers, while sparse private parking and constrained local budgets exacerbate reliance on public infrastructure (Mattioli, 2014, Stacey and Shaddock, 2015). These areas also tend to exhibit higher rates of transport poverty and social exclusion, meaning that without targeted interventions the roll-out of public chargepoints risks deepening existing inequalities rather than ameliorating them (Carroll et al., 2021). From an energy-justice perspective, ensuring that the benefits of decarbonisation accrue to those most in need - and that no community is left behind - demands explicit focus on historically deprived neighbourhoods (Sovacool and Dworkin, 2015).

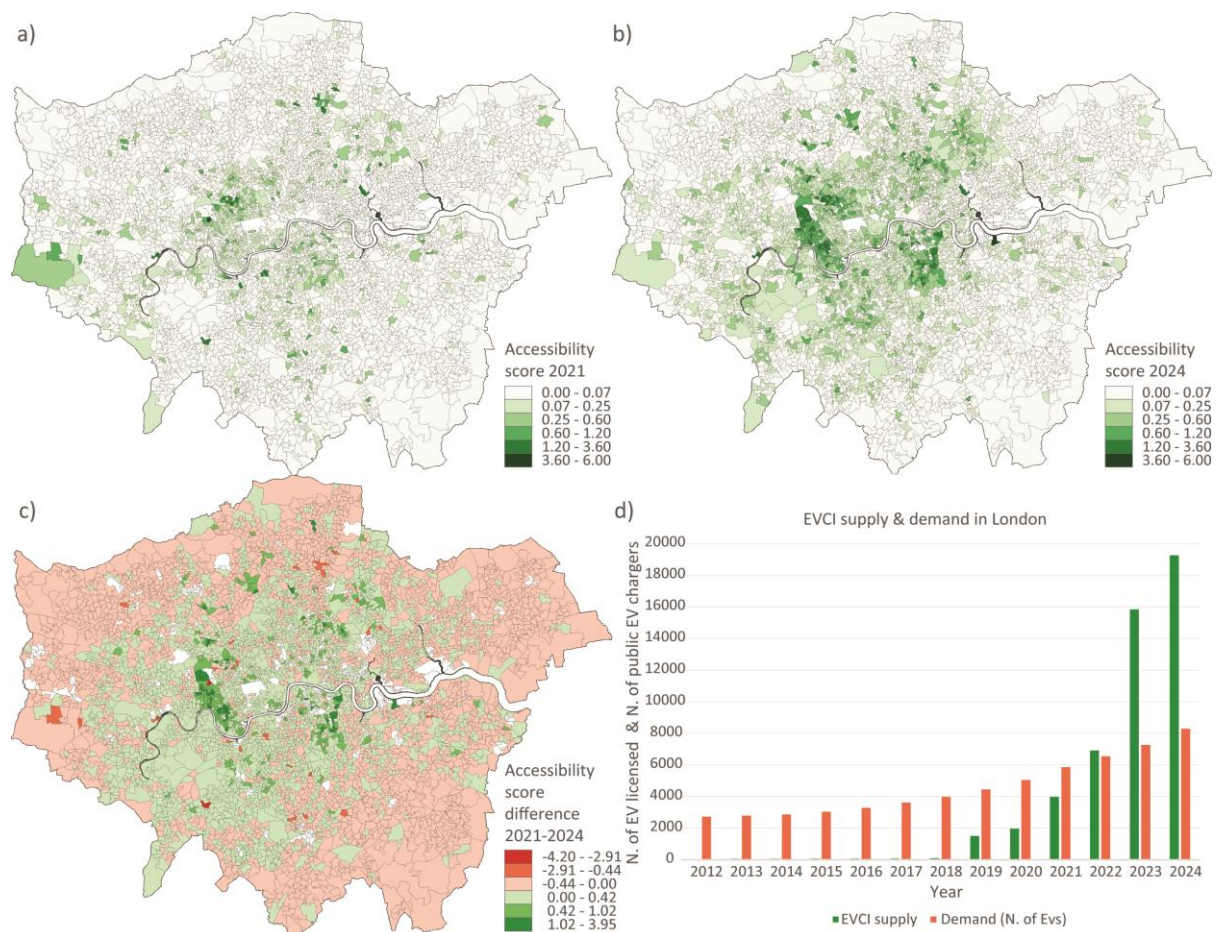


Figure 1 a) Public EVCI accessibility score in 2021 - LSOA level. b) Public EVCI accessibility score in 2024 - LSOA level. c) Difference in public EVCI accessibility score between 2021 and 2024 - LSOA level. d) Number of licensed Electric Vehicles and public charging points in London from 2012 to 2024.

Figure 1.a and Figure 1.b show how the public charging infrastructure development between 2021

and 2024 is not spatially uniform and does not uniformly address the increase in demand. We defined an accessibility metric as ratio of supply of public chargepoints over demand (represented by licensed EVs per zone), at the spatial level of Lower Super Output Area (LSOA). The formulation is the following:

$$A_j = \frac{EVCI_j}{EVL_j} \quad (1)$$

Where, A_j is the accessibility to public EVCI at location j , $EVCI_j$ is the number of public EVCI at location j , and EVL_j is the number of licenced EVs at location j .

To evaluate changes over time, we calculated the difference between two years - see equation (2). This metric allowed us to assess how accessibility has evolved spatially and temporally.

$$A_j^{diff} = A_j^{t2} - A_j^{t1} \quad (2)$$

Where A_j^{t2} is the public EVCI accessibility score considering demand and supply in 2024 ($t2 = 2024$), A_j^{t1} is the public EVCI accessibility score considering demand and supply in 2021 ($t1 = 2021$), and A_j^{diff} is the difference in public EVCI accessibility.

The Gini index for 2021 public EVCI accessibility at LSOA level has a value of 0.81, while for 2024 it has a value of 0.58, indicating that inequalities have decreased, but still persist and significant. As shown in Figure 1.c, despite the faster increase of EVCI supply with respect to demand, several areas failed to translate into equitable accessibility gains: the red areas in the map actually saw a decrease in the accessibility score in 2024 with respect to 2021.

Apart from accessibility difference, we also calculated the absolute number of public EVCI supply variation (i.e., increased/decreased number of public EVCI per zone: $SV_j = S_j^{t2} - S_j^{t1}$) and demand variation (i.e., increased/decreased number of EV licensing per zone: $DV_j = D_j^{t2} - D_j^{t1}$) to better explain where the above-mentioned inequalities come from.

To analyse the factors shaping these trends, the indices we calculated, including A_j^{t1} , A_j^{t2} , A_j^{diff} , SV_j and DV_j are used as dependent variables. We first applied both univariate and multivariate Ordinary Least Squares (OLS) regressions and a Spatial Autoregressive (SAR) model, to identify statistically significant drivers of the identified accessibility difference between 2021 and 2024 at the global level and Geographically Weighted Regression (GWR) to uncover spatial variations.

Table 1 Data and sources

Data	Description	Source
EVCI supply	Number of public EV chargepoint per LSOA	Department for Transport. National charge point registry.
EVCI demand	Number of EV licensed per LSOA	Department for Transport and Driver and Vehicle Licensing Agency. Vehicle licensing statistics data tables.
Approximated Social Grade (ASG)	Number of ASG individuals in different categories per LSOA. AB_C1: Higher and intermediate managerial/administrative/professional occupations (AB), and Supervisory, clerical, and junior managerial/administrative/professional occupations (C1) C2_DE: Skilled manual occupations (C2) and Semi-skilled and unskilled manual occupations; unemployed and lowest grade occupations (DE)	Office for National Statistics (ONS), NOMIS SG002EW, Approximated Social Grade. Census 2021.
Deprivation	Number of households (HH) deprived in 3 or more dimensions (D3+) per LSOA.	Office for National Statistics (ONS), Households by deprivation dimensions. Census 2021.
Car ownership	Number of households with either: 1) zero or one car, or 2) two or more vehicles.	Office for National Statistics (ONS), Car or van availability. Census 2021.
House tenure	Number of households living in each different category: owned and rented.	Office for National Statistics (ONS), Tenure. Census 2021.
Accommodation type	Number of households living in flats, terraced, detached & semidetached, or other kind of accommodations.	Office for National Statistics (ONS), Accommodation type. Census 2021.
House prices	Median price paid for residential properties by LSOA.	Office for National Statistics (ONS), HPSSA Dataset 46.
Population density	Population Density for Lower Layer Super Output Areas	Office for National Statistics (ONS), Table SAPE22DT11
POI density	Points of interest from OpenStreetMap.	OpenStreetMap. https://www.openstreetmap.org

Based on the literature on the main factors influencing EV adoption (Franke and Krems, 2013, Coffman et al., 2017, Vassileva and Campillo, 2017, Chen et al., 2020a, Ling et al., 2021), we considered a suite of socio-demographic and urban form variables (Table 2), selected to capture the key dimensions shaping public EVCI accessibility across London LSOAs. These include measures of social grade (s-ASG_ABC1, s-ASG_C2DE), deprivation (s-D3+), car-ownership patterns (s-HHcars01, s-HHcars2+), household tenure (s-HHT_own, s-HHT_rent), and accommodation typologies (s-semi-det, s-flat, s-terr-oth), as well as market and urban structure indicators - median house prices in 2021 and 2023 (Med_HP_21, Med_HP_23), population density (Pop_dens) and points-of-interest density (POI_dens). We also explored additional predictors, like different measures of road network density and local jobs distribution, but these never reached statistical significance in any of our regression specifications and were therefore omitted from the final models.

Table 2 Variables used in the regression analyses

Category	Variable	Description
Approximated Social Grade (ASG)	s-ASG_ABC1	Normalised LSOA share of individuals with higher ASG: categories A, B and C1.
	s-ASG_C2DE	Normalised LSOA share of individuals with lower ASG: categories C2, D, and E.
Deprivation	s-D3+	Normalised LSOA share of households deprived in 3 or more dimensions.
Car ownership	s-HHcars01	Normalised LSOA share of households with 0 or 1 car.
	s-HHcars2+	Normalised LSOA share of households with 2 or more cars.
Household tenure (HHT)	s-HHT_own	Normalised LSOA share of households who own their accommodation.
	s-HHT_rent	Normalised LSOA share of households renting accommodation.
Accommodation type	s-semi-det	Normalised LSOA share of detached and semidetached houses.
	s-flat	Normalised LSOA share of flats.
	s-terr-oth	Normalised LSOA share of terraced and other houses.
House prices	Med_HP_21	LSOA median house prices in 2021
	Med_HP_23	LSOA median house prices in 2023
Urban features	Pop_dens	LSOA population density
	POI_dens	LSOA POI density

To unpack the drivers of EVCI accessibility at the metropolitan scale, we first ran univariate OLS regressions (equation (3)) for each independent variable, isolating its raw association with the accessibility score, before progressing to multivariate OLS to control for confounding and quantify net effects. The univariate regression model takes the following general form:

$$y_i = \alpha + \beta X_i + \epsilon_i \quad (3)$$

Where $y_i \in Y = [y_1, y_2, y_3, y_4, y_5]$, Y is the set of the dependent variables ($A_j^{t1}, A_j^{t2}, A_j^{diff}, SV_j, DV_j$), α is the intercept, $X_i = [s\text{-ASG_ABC1}, s\text{-ASG_C2DE}, s\text{-D3+}, s\text{-HHcars01}, s\text{-HHcars2+}, s\text{-HHT_own}, s\text{-HHT_rent}, s\text{-semi-det}, s\text{-flat}, s\text{-terr-oth}, \text{Med_HP_21}, \text{Med_HP_23}, \text{Pop_dens}, \text{POI_dens}]$ is the vector of the independent variables (described in Table 2), and ϵ_i is the idiosyncratic error term. We estimate equation (3) for each combination of dependent and independent variables.

We then run a multivariate OLS regression (equation (4)) introducing controls.

$$y_i = \gamma + \delta X'_i + u_i \quad (4)$$

Where $y_i \in Y = [y_1, y_2, y_3, y_4, y_5]$, Y is the set of dependent variables ($A_j^{t1}, A_j^{t2}, A_j^{diff}, SV_j, DV_j$), γ is the intercept, $X'_i = [s\text{-ASG_C2DE}, s\text{-D3+}, s\text{-HHcars01}, s\text{-HHT_rent}, s\text{-flat}, s\text{-terr-oth}, \text{Med_HP_21}, \text{Med_HP_23}, \text{Pop_dens}, \text{POI_dens}]$ is the vector of the independent variables (described in Table 2), and u_i is the idiosyncratic error term.

Recognising that spatial clustering may bias standard OLS estimates, we then implemented a Spatial Autoregressive (SAR) model (equation (5)), which explicitly incorporates a spatial-lag term to capture spillovers between neighbouring LSOAs as in reality, people drive to wherever EVCI is available to charge.

$$y_i = \rho W y_i + \theta X'_i + e_i \quad (5)$$

Where W is the row-standardised spatial weights matrix capturing adjacency of wards, ρ the spatial-lag parameter measuring the strength of spillovers, θ is the vector of coefficients, and e is the vector of residuals.

Finally, to reveal local heterogeneity in these relationships, we applied Geographically Weighted Regression (GWR), allowing coefficients to vary across space and thus uncovering neighbourhood-level variations in how socio-spatial factors influence accessibility.

$$y_i = \beta(u_i, v_i)X_i + \varepsilon_i \quad (6)$$

Where each location i at coordinates (u_i, v_i) has its own coefficient vector $\beta(u_i, v_i)$, estimated by weighting nearby observations more heavily (via a Gaussian kernel). GWR thus produces spatially varying coefficient surfaces and local goodness-of-fit (local R^2) maps, which we interpret to identify wards where particular predictors exert stronger or weaker influences - revealing pockets of heterogeneity that global models mask.

The choice of this multi-model strategy rests on balancing global inference with spatial nuance. Univariate OLS provides straightforward effect sizes but ignores covariation; multivariate OLS adjusts for multiple predictors yet assumes spatial independence; SAR corrects for spatial autocorrelation but still yields a single “average” coefficient surface; while GWR relaxes the stationarity assumption entirely, mapping how each driver’s influence waxes and wanes across London. Together, these complementary models offer both robust, city-wide insights and fine-grained, place-specific diagnostics, for our findings to represent the complex geography of EVCI equity.

3. Results

3.1 OLS Analysis results: uneven patterns of accessibility and infrastructure growth

The analysis of public EVCI accessibility in 2021 (A_j^{t1}), in 2024 (A_j^{t2}), the differences between 2021 and 2024 A_j^{diff} , and the supply (SV_j) & demand (DV_j) variations between 2021 and 2024 highlights several significant correlations (Table 3).

The univariate OLS results (Table 3) reveal that socio-economic status is strongly associated with patterns of EV charging point accessibility and its evolution between 2021 and 2024. In particular, areas characterised by higher social grades (ASG_ABC1) exhibit significantly positive associations with accessibility scores in 2024, as well as with the observed increase in accessibility over the 2021-2024 period (i.e., positive values of A_j^{diff}). Conversely, neighbourhoods with lower social grades (ASG_C2DE) show negative relationships with both accessibility levels and accessibility gains. This divergence is mirrored in the changes in public EVCI supply and demand: higher social grade areas experience larger increases in both the number of public charging points and licensed electric vehicles, whereas lower social grades are negatively correlated with such growth. These patterns suggest that, without intervention, the rollout of charging infrastructure and EV uptake risk reinforcing existing socioeconomic hierarchies.

Household deprivation (measured as households deprived in three or more dimensions) exhibits a more nuanced picture. Although it correlates positively with accessibility scores in 2021 and 2024, and with accessibility gains over time - likely reflecting higher population density in deprived areas (Figure 2) - it nonetheless correlates negatively with increases in both EVCI supply and EV licensing. In other words, while deprived neighbourhoods may already display relatively high baseline accessibility (driven in part by dense urbanisation), they have not seen commensurate growth in public charging infrastructure or EV ownership. A similar density effect (see also correlation among variables in Figure 2) is evident in household car ownership patterns: zones with predominantly zero- or one-car households demonstrate positive correlations with accessibility in both survey years and with EVCI supply increases, yet correlate negatively with EV registration growth. In contrast, areas with two or more cars recorded the greatest rise in licensed EVs. This suggests that, despite relatively poorer accessibility to public EVCI in higher car-owning areas, it is precisely these better-resourced households that drive EV uptake.

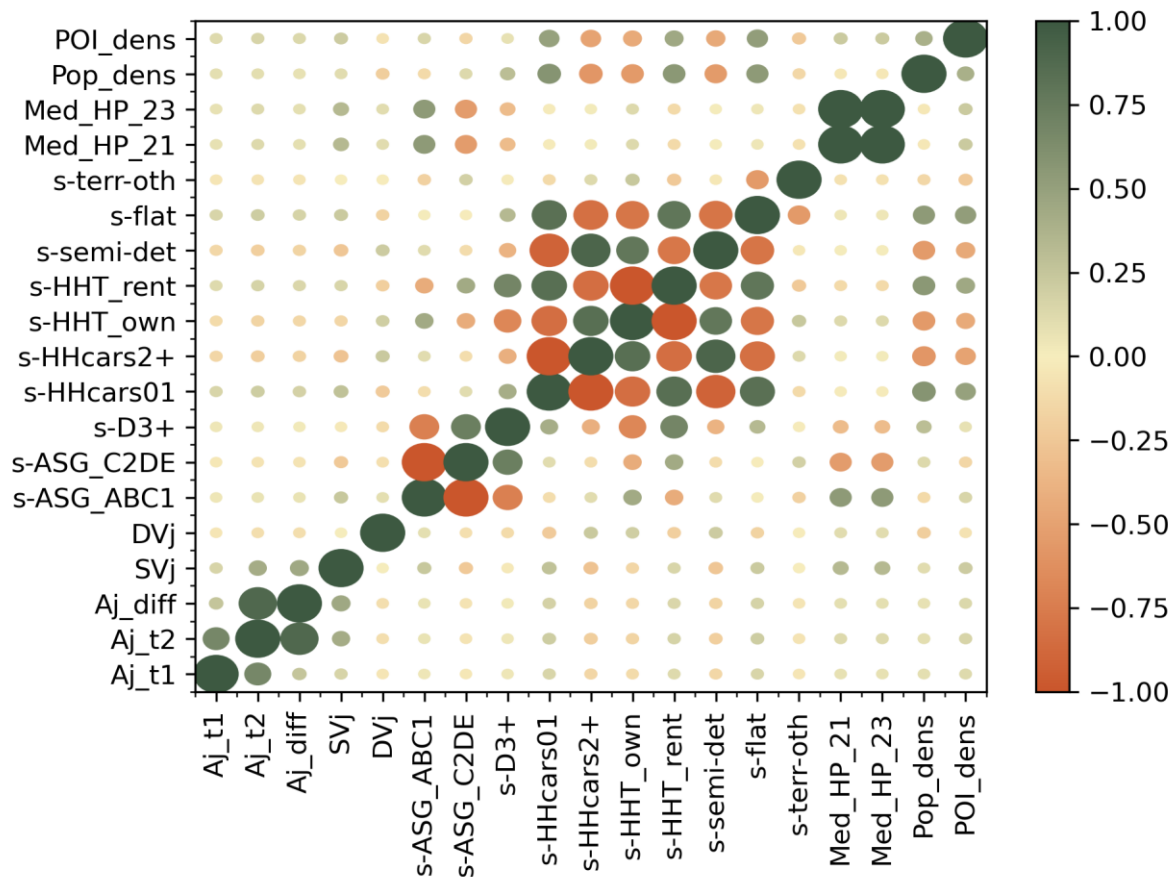


Figure 2 Correlation matrix.

Tenure and housing typology further differentiate how neighbourhoods have evolved. Owner-occupied areas display a positive relationship only with increases in licensed electric vehicles, and negative correlations with all other variables (accessibility, EVCI supply, and demand growth). In contrast, neighbourhoods with a high proportion of renters exhibit a negative association with accessibility in 2021, which reverses to a positive relationship in 2024, signalling a shift in where charging improvements have occurred. Renters also show a positive correlation with increases in EVCI supply - again likely a density artefact - but a negative link to EV licensing increases. Detached and semi-detached housing types are negatively correlated with accessibility and EVCI supply, indicating that more suburban, lower-density neighbourhoods are comparatively underserved by public charge points, likely due to private charging behaviours. Flat-dominated areas, by contrast, have higher public EVCI accessibility and more rapid growth in public EVCI supply, although they see smaller rises in EV licensing. Finally, local market conditions (median house prices in 2021 and 2023) are consistently positively correlated with increases in accessibility, supply, and demand, while population density aligns positively with accessibility and EVCI supply growth but negatively with EV licensing growth. Points-of-interest density shows weak or non-significant associations with most outcomes.

When these variables are introduced jointly in a multivariate OLS framework (Table 4) - controlling for compositional interdependencies and removing perfectly collinear proportions - the core patterns largely persist. Lower social grade remains a significant negative predictor of both accessibility improvement and EVCI supply increase, confirming that areas with more economically disadvantaged residents are falling behind in charging infrastructure provision. Households with zero or one car continue to be positively associated with accessibility gains and supply expansion, yet negatively associated with demand growth, indicating that electrification is progressing most rapidly among larger, better-endowed households. Renting tenure emerges as a positive predictor of accessibility

increase, reflecting targeted infrastructure rollout in denser, more rental-dominated sections of the city. Median house price retains its positive association with all the dependent variables underscoring the link between local affluence and EV transition dynamics. Population density remains positively associated with accessibility gains yet negatively linked to EV licensing growth, reinforcing the earlier inference that dense urban neighbourhoods are improving in public charging availability but lag in EV uptake relative to less dense areas. POI density fails to achieve statistical significance in most specifications. Collectively, these multivariate findings underscore the persistence of socio-spatial inequalities: wealthier, higher-status areas continue to benefit more from both charging infrastructure provision and EV adoption, whereas deprived or less affluent communities - despite relatively high baseline accessibility - have experienced fewer tangible gains in supply and demand growth.

Table 3 Univariate OLS results

Description	Dep. variable	(1) A_j^{t1}	(2) A_j^{t2}	(3) A_j^{diff}	(4) SV_j	(5) DV_j
High social grades	s-ASG_ABC1	-0.002 (0.004)	0.032*** (0.005)	0.015*** (0.002)	0.077*** (0.052)	0.008*** (0.001)
Low social grades	s-ASG_C2DE	-0.005 (0.004)	-0.032*** (0.005)	-0.015*** (0.002)	-0.077*** (0.005)	-0.008*** (0.001)
Deprivation	s-D3+	0.021** (0.010)	0.050*** (0.007)	0.013*** (0.004)	-0.015* (0.008)	-0.018*** (0.002)
HH w/ 0 or 1 car	s-HHcars01	0.013** (0.006)	0.139*** (0.005)	0.050*** (0.003)	0.113*** (0.005)	-0.022*** (0.002)
HH w/ 2+ cars	s-HHcars2+	-0.020*** (0.006)	-0.139*** (0.005)	-0.050*** (0.003)	-0.113*** (0.005)	0.022*** (0.002)
Homeowners	s-HHT_own	0.004 (0.005)	-0.086*** (0.004)	-0.030** (0.002)	-0.054*** (0.004)	0.015*** (0.001)
Renters	s-HHT_rent	-0.010* (0.006)	0.086*** (0.004)	0.030*** (0.002)	0.054*** (0.004)	-0.015*** (0.001)
(Semi)detached h.	s-semi-det	-0.010** (0.005)	-0.094*** (0.004)	-0.033*** (0.002)	-0.079*** (0.004)	0.015*** (0.001)
Flats	s-flat	0.004 (0.004)	0.077*** (0.003)	0.027*** (0.002)	0.061*** (0.003)	-0.010*** (0.001)
Terraced/other h.	s-terr-oth	-0.001 (0.003)	-0.016*** (0.005)	-0.006** (0.003)	-0.004*** (0.005)	-0.004*** (0.001)
House prices ('21)	Med_HP_21	0.039*** (0.009)	-	-	-	-
House prices ('23)	Med_HP_23	-	0.097*** (0.009)	0.037*** (0.005)	0.216*** (0.009)	0.019*** (0.003)
Population density	Pop_dens	0.037*** (0.008)	0.177*** (0.008)	0.061*** (0.004)	0.104*** (0.008)	-0.033*** (0.002)
POI density	POI_dens	0.004 (0.009)	0.162*** (0.010)	0.059*** (0.005)	0.171*** (0.010)	-0.010*** (0.003)
R2		0.061	0.011	0.009	0.060	0.009

The estimation model is univariate OLS. Each row represents an estimated coefficient from a univariate regression of the dependent variable indicated in the column header against the variable indicated in the rows. Standard errors are presented in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4 Multivariate OLS results

Description	Dep. variable	(1) A_j^{t1}	(2) A_j^{t2}	(3) A_j^{diff}	(4) SV_j	(5) DV_j
High social grades	s-ASG_ABC1	-	-	-	-	-
Low social grades	s-ASG_C2DE	-0.004 (0.007)	-0.045*** (0.009)	-0.024*** (0.005)	-0.037*** (0.009)	0.000 (0.003)
Deprivation	s-D3+	0.021** (0.010)	0.025** (0.012)	0.003 (0.007)	0.009 (0.012)	-0.002 (0.004)
HH w/ 0 or 1 car	s-HHcars01	0.032*** (0.012)	0.095*** (0.015)	0.038*** (0.008)	0.093*** (0.015)	-0.023*** (0.005)
HH w/ 2+ cars	s-HHcars2+	-	-	-	-	-
Homeowners	s-HHT_own	-	-	-	-	-
Renters	s-HHT_rent	-0.014 (0.011)	0.018 (0.014)	0.017** (0.008)	0.017 (0.013)	0.007* (0.004)
(Semi)detached h.	s-semi-det	-	-	-	-	-
Flats	s-flat	0.015* (0.008)	-0.005 (0.011)	-0.011* (0.006)	0.006 (0.010)	0.001 (0.03)
Terraced/other h.	s-terr-oth	0.009 (0.006)	0.007 (0.008)	0.000 (0.004)	0.029*** (0.008)	-0.002 (0.02)
House prices ('21)	Med_HP_21	0.039*** (0.009)	-	-	-	-
House prices ('23)	Med_HP_23	-	0.069*** (0.012)	0.020*** (0.006)	0.188*** (0.012)	0.018*** (0.004)
Population density	Pop_dens	0.037*** (0.008)	0.075*** (0.010)	0.024*** (0.006)	-0.006 (0.010)	-0.022*** (0.003)
POI density	POI_dens	0.004 (0.009)	0.001 (0.011)	-0.002 (0.006)	0.014 (0.011)	0.006* (0.004)
R2		0.061	0.192	0.094	0.233	0.066

The estimation model is multivariate OLS. Standard errors are presented in parentheses.
 * p < 0.1, ** p < 0.05, *** p < 0.01

3.2 Findings from the SAR model: accounting for spatial dependence

To assess the presence and nature of spatial dependence in the data, we conducted a series of Lagrange Multiplier (LM) tests: the LM-Lag and LM-Error tests, along with their respective robust versions. The standard LM tests detect whether spatial autocorrelation exists in the lagged dependent variable (LM-Lag) or in the error terms (LM-Error). However, since the presence of one form of spatial dependence can inflate the statistic of the other, the robust versions help isolate the dominant source of spatial autocorrelation. In this analysis, the LM-Lag and Robust LM-Lag tests were statistically significant across most dependent variables, suggesting that spatial dependence operates primarily through the lagged dependent variable rather than through spatially correlated errors. These results support the use of a Spatial Autoregressive (SAR) model, which accounts for the influence that neighbouring observations exert on the dependent variable.

Table 5 Lagrange Multiplier (LM) tests results

Dep. variable	(1) A_j^{t1}	(2) A_j^{t2}	(3) A_j^{diff}	(4) SV_j	(5) DV_j
LM lag	405.8*	1774.8*	959.5*	2129.6*	5.8*
LM error	390.8*	1706.4*	918.0*	2158.4*	4.5*
Robust LM lag	16.1*	73.7*	43.0*	41.0*	2.6
Robust LM error	1.1	5.3*	1.5	69.8*	1.3

* $p < 0.05$

Accounting for spatial dependence through a Spatial Autoregressive (SAR) framework alters several associations observed under OLS and highlights the role of neighbourhood spillovers in shaping public EVCI accessibility equity. In the SAR model, lower social grades (ASG_C2DE) retain a significant, negative association with public EVCI accessibility in 2024. Moreover, the lagged spatial term accentuates that improvements in accessibility between 2021 and 2024 are similarly suppressed in lower ASG areas, as are increases in public charging point supply. In practical terms, neighbourhoods characterised by a higher concentration of lower status households not only experience poorer charging access themselves but also exert a dampening effect on adjacent areas, reinforcing socio-economic divides across contiguous LSOAs.

Contrary to its positive univariate relationship, household deprivation (s-D3+) becomes statistically insignificant once spatial lag effects are introduced. This shift suggests that the apparent link between deprivation (defined as specified in Table 2) and accessibility under OLS is largely an artefact of clustering among deprived areas, rather than a direct driver of infrastructure provision or accessibility scores. In other words, when one accounts for how deprivation clusters spatially, there is no residual net effect on EVCI outcomes; deprived neighbourhoods neither systematically attract nor repel charging infrastructure once their neighbours' characteristics are controlled for. The absence of a direct deprivation effect underlines the importance of disentangling first-order spatial clustering from substantive socio-economic influences.

Household car ownership patterns remain relevant under spatial modelling. Areas dominated by zero- or one-car households maintain a positive correlation with public EVCI accessibility in both 2021 and 2024, and with improvements in accessibility over time. They also exhibit a positive link to growth in public charging point stock. However, the SAR specification confirms a strong, negative association between one-car (or carless) areas and EV adoption growth, indicating that neighbourhoods where multi-car households predominate drive the increase in licensed electric vehicles. This juxtaposition - where neighbourhoods with fewer cars benefit most from access improvements but lag in uptake - persists even after accounting for spatial spillovers, emphasising the continued divide between areas optimised for public infrastructure and those leading in EV ownership.

Tenure effects are partly attenuated in the presence of spatial spillovers. Although OLS identified a positive connection between renting tenure and supply increases, the SAR model finds that the proportion of renters in an LSOA loses statistical significance with respect to both accessibility and public EVCI supply once neighbouring characteristics are incorporated. The only tenure-related effect to remain significant is a strong, negative relationship between renting and EV licensing growth, suggesting that regions with high owner-occupation rates continue to spearhead EV adoption. Flat-dominant neighbourhoods, which under univariate tests correlated positively with accessibility and supply, now reveal a negative relationship with accessibility improvements between 2021 and 2024. Conversely, areas with a greater share of terraced housing are positively linked to public EVCI supply growth. These nuanced shifts illustrate how the SAR framework can uncover that some characteristics - like tenure composition - exert influence largely through spatial clustering rather than as independent predictors.

Market dynamics and amenity density exhibit further attenuation under spatial controls. Median house price remains positively associated with EVCI accessibility in 2021, but its links to supply and demand increases become markedly weaker when spatial lag effects are taken into account. This weakening implies that observed associations between affluence and infrastructure or EV uptake are partly explained by the wider residential context, whereby high-value neighbourhoods tend to cluster together. Finally, POI density emerges only as a weak, positive predictor of EV adoption growth, with no significant relationship to accessibility or supply once spatial dependence is modelled. Collectively, the SAR results underscore that, while socio-economic and built environment factors do matter, their effects are deeply embedded in and modulated by the spatial structure of London's neighbourhoods.

Table 6 Spatial Autoregressive (SAR) model results

Description	Dep. variable	(1) A_j^{t1}	(2) A_j^{t2}	(3) A_j^{diff}	(4) SV_j	(5) DV_j
High social grades	s-ASG_ABC1	-	-	-	-	-
Low social grades	s-ASG_C2DE	-0.027	-0.143***	-0.162***	-2.474***	-4.992
Deprivation	s-D3+	0.250	0.247	0.111	5.922	-5.453
HH w/ 0 or 1 car	s-HHcars01	0.119**	0.287***	0.224***	8.384***	-39.609***
HH w/ 2+ cars	s-HHcars2+	-	-	-	-	-
Homeowners	s-HHT_own	-	-	-	-	-
Renters	s-HHT_rent	-0.053	0.016	0.087	-0.676	-11.863*
(Semi)detached h.	s-semi-det	-	-	-	-	-
Flats	s-flat	0.041	-0.023	-0.068*	-1.010	-0.452
Terraced/other h	s-terr-oth	0.033	0.030	-0.005	1.249**	-4.298
House prices ('21)	Med_HP_21	0.00000002**	-	-	-	-
House prices ('23)	Med_HP_23	-	0.000	0.000	0.000002**	0.00001***
Population density	Pop_dens	0.000002***	0.000002***	0.000001*	-0.00004***	-0.001***
POI density	POI_dens	0.000	0.000	0.000	0.000	0.005*
	rho	0.372	0.616	0.505	0.642	0.059
	Log_likelihood	1582.3	624.7	538.0	-12345.0	40574
	AIC	-3142.6	-1227.5	-1054.0	24782.1	40574.0
	R2	0.15	0.44	0.27	0.50	0.07

The estimation model is SAR.

* p < 0.1, ** p < 0.05, *** p < 0.01

3.3 Patterns observed through GWR analysis: variabilities across locations

The GWR model was employed to explore the spatial variation in the relationships between the same range of dependent and independent variables explored in the previous analyses. The spatial distribution of local R^2 values is presented in Figure 3, while the spatial patterns of the local coefficients are shown in Figure 4 through Figure 7. Areas where coefficients are not statistically significant (based on local t-values) are greyed out in the maps.

Figure 4 focuses on a subset of independent variables that are statistically significant in both EVCI supply and demand models, allowing for a comparative analysis of key drivers behind infrastructure deployment and EV uptake. These results show that public EVCI supply is negatively correlated with the share of households owning 0 or 1 cars in central London, but positively correlated with the same variable in west and central-east London. In contrast, EV adoption exhibits a citywide negative correlation with car ownership, strongest in the north-east. Public EVCI accessibility in 2021 shows a positive correlation with car ownership in east London, and this pattern strengthens in 2024 -

particularly in both east and west London - though it weakens in central and south London. A similar pattern (but with lower coefficients) appears when looking at accessibility improvements between 2021 and 2024.

Figure 5 through Figure 7 present the spatial variation in coefficients for each dependent variable grouped by thematic categories. Socio-economic indicators (Figure 5) reveal that areas with lower approximated social grades (ASG) tend to have lower public EVCI accessibility across 2021, 2024, and the period change, particularly in the southwest, northeast, and parts of west and south London. Deprivation (D3+) shows a more variable pattern, with positive associations in west London (2021), central London (2024), and some improvement in accessibility in east London, though west London shows a negative correlation over time.

Housing-related variables (Figure 6) highlight that areas with higher renters' shares are positively associated with public EVCI accessibility in 2024 and its increase in west London, while zones with higher flats' shares show the inverse relationship in the same areas. Terraced housing is rarely statistically significant, although a weak positive relationship with accessibility in 2024 appears in central-south and east London.

Figure 7 focuses on market and city structure indicators. Median house prices are consistently and strongly positively correlated with EVCI supply increases and EV adoption across most of the city, with exceptions in some eastern and northern boroughs. Accessibility in 2021 and 2024 also correlate positively with house prices, especially in north-west and south-east London, although some negative correlations emerge in central-west London in 2024. Population density has a mixed effect: positively associated with EVCI supply only in west London, and negatively associated with EV adoption citywide. However, it shows a consistent positive relationship with EVCI accessibility across all three temporal measures. POI density rarely yields statistically significant results but shows a few localised effects - negative in west and south London (Chiswick and Southwark) for supply, weakly positive in central London for EV adoption, and weakly negative in central areas for accessibility in 2024 and its improvement over time.

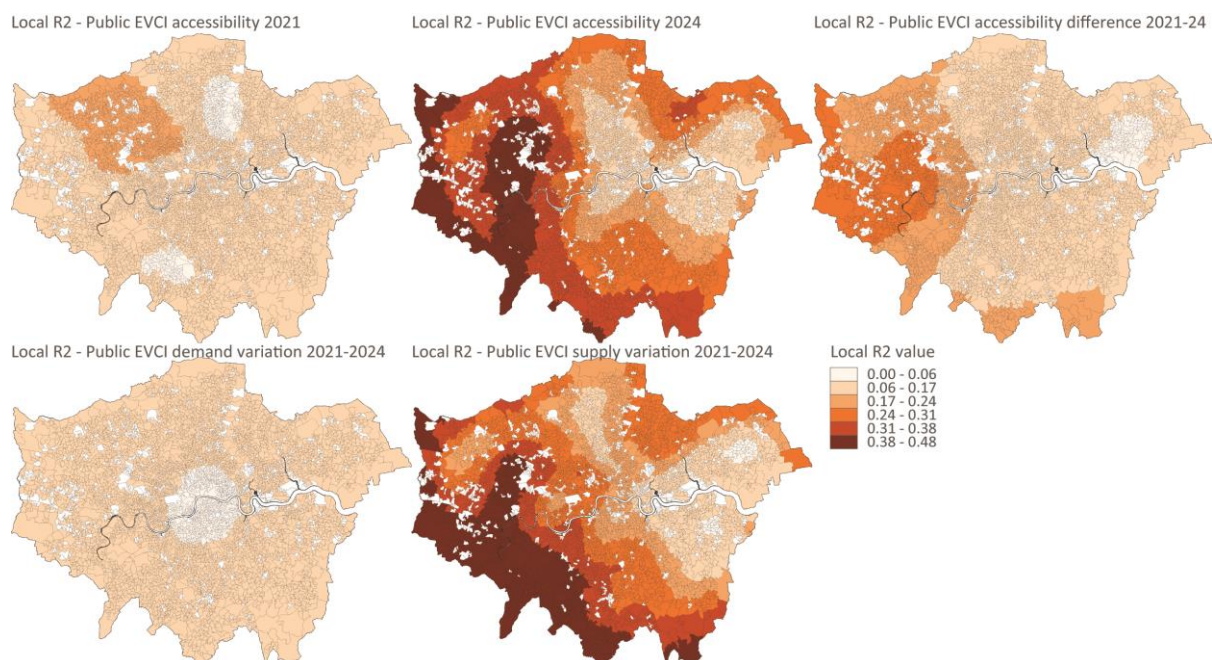
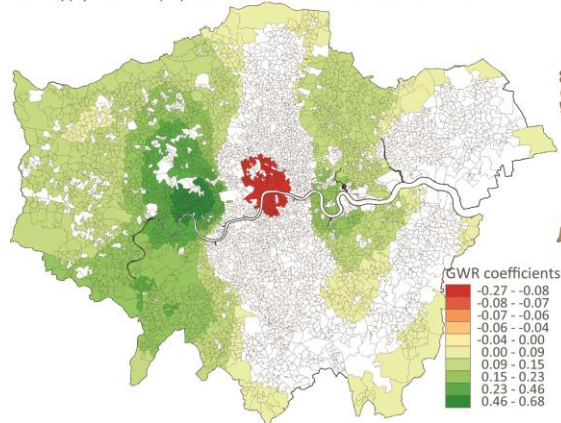
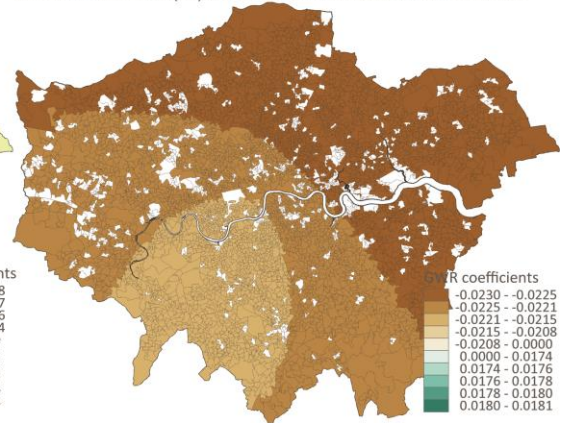


Figure 3 Local R2 values for public EVCI accessibility in 2021, 2024, the difference between 2021 and 2024, and the increase of EVCI demand and supply between 2021 and 2024.

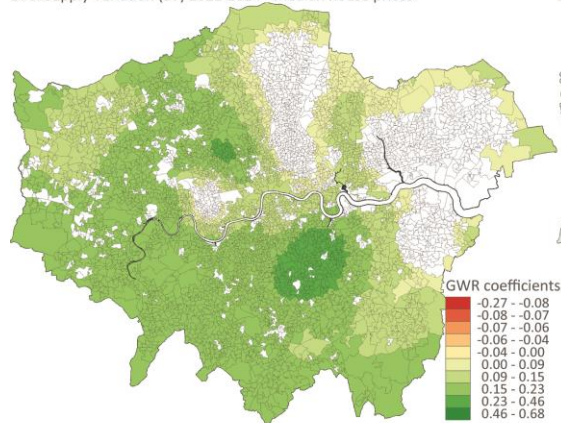
EVCI Supply Variation (SV) 2021-2024 - Share of households with 0 or 1 car



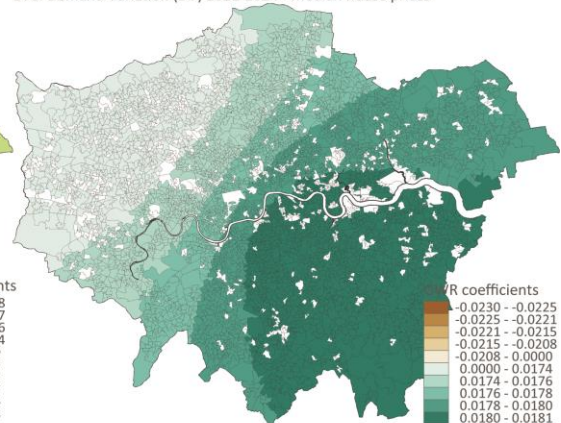
EVCI Demand Variation (DV) 2021-2024 - Share of households with 0 or 1 car



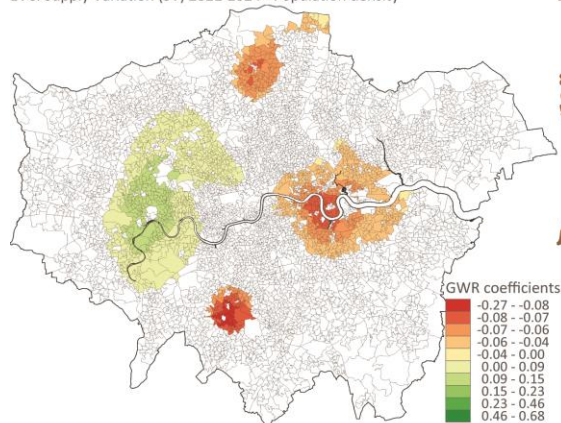
EVCI Supply Variation (SV) 2021-2024 - Median house prices



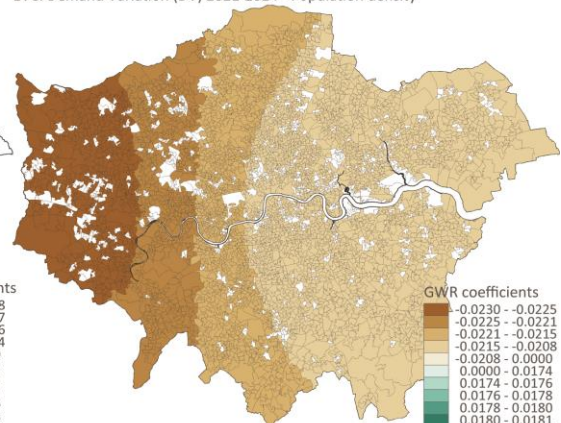
EVCI Demand Variation (DV) 2021-2024 - Median house prices



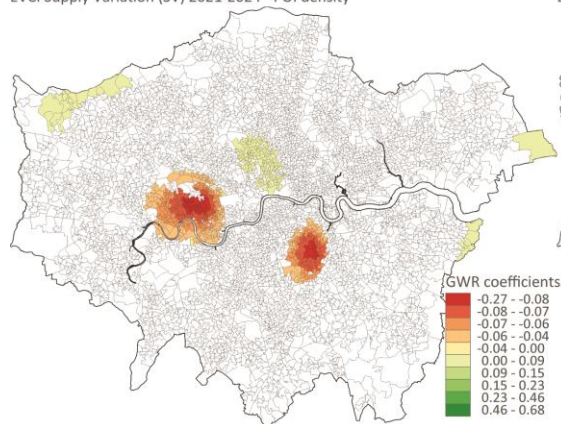
EVCI Supply Variation (SV) 2021-2024 - Population density



EVCI Demand Variation (DV) 2021-2024 - Population density



EVCI Supply Variation (SV) 2021-2024 - POI density



EVCI Demand Variation (DV) 2021-2024 - POI density

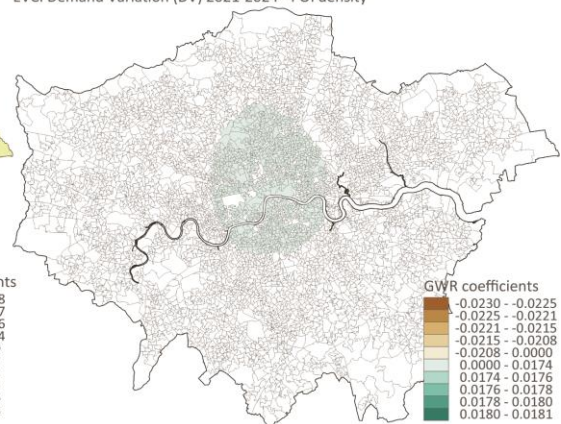


Figure 4 GWR coefficients of EVCI supply and demand increase between 2021 and 2024.

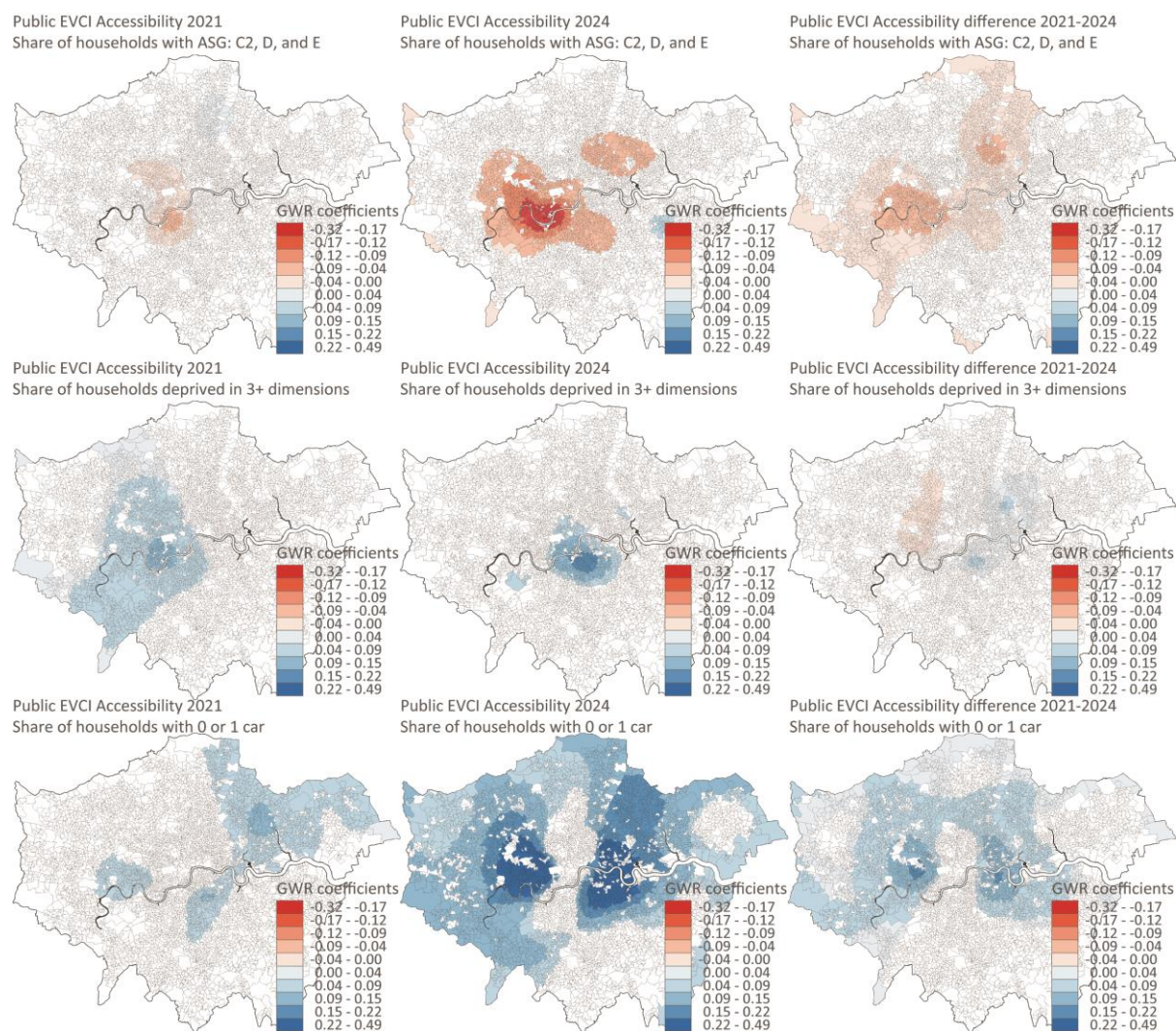


Figure 5 GWR coefficients of Approximated Social Grade (ASG), deprivation, and car ownership in 2021, 2024, and the difference between 2021 and 2024.

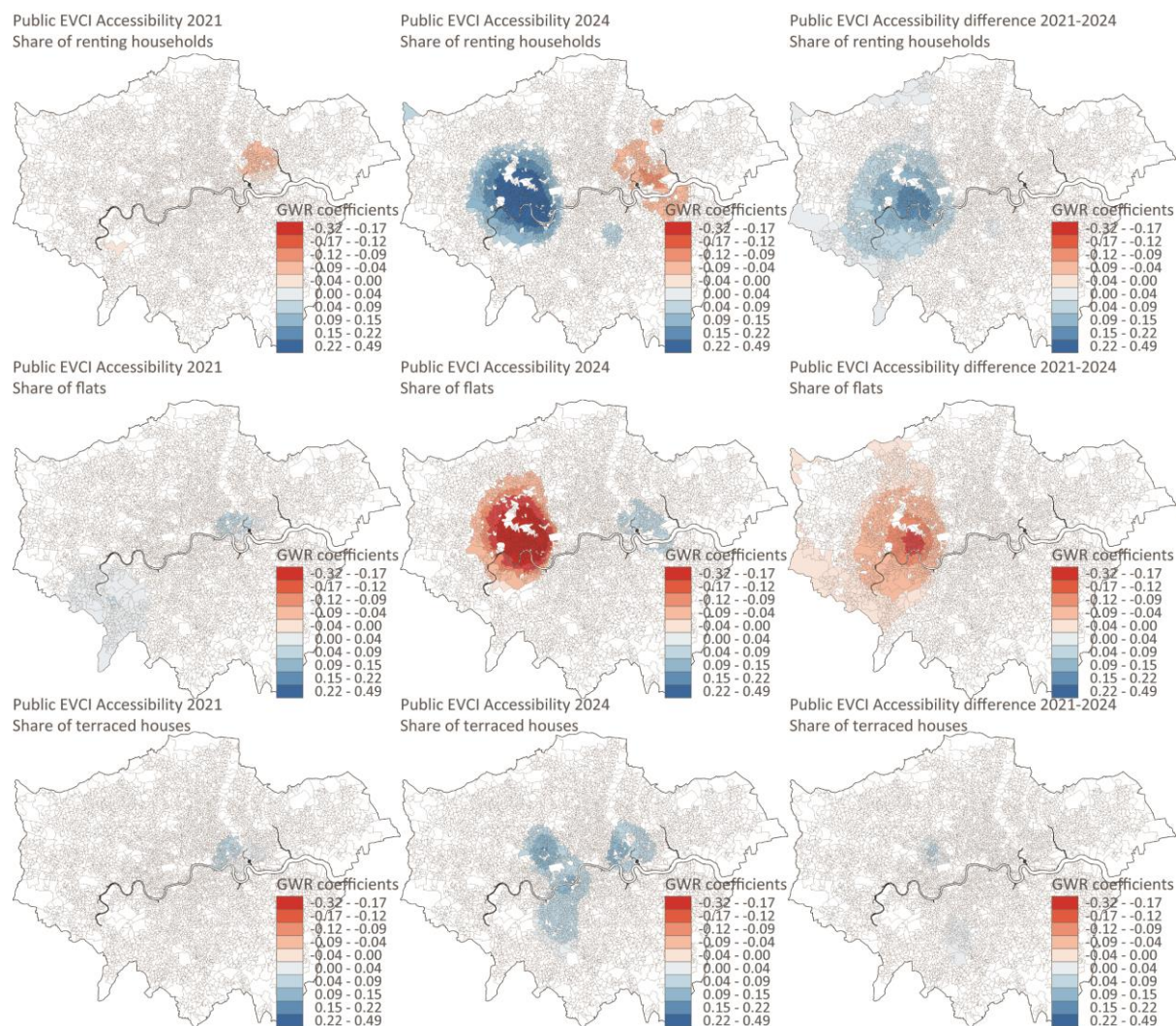


Figure 6 GWR coefficients of different accommodation types and house tenure in 2021, 2024, and the difference between 2021 and 2024.

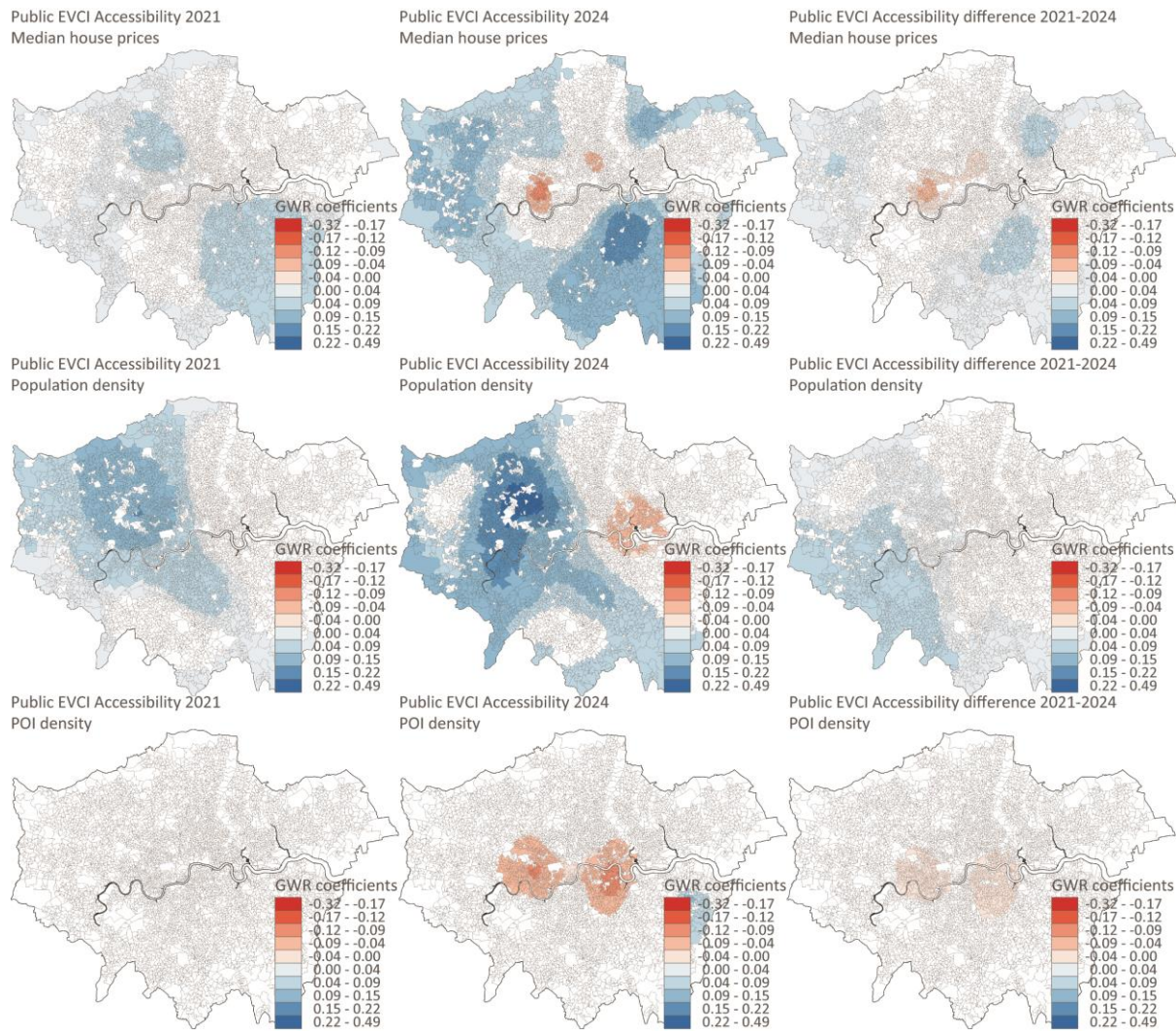


Figure 7 GWR coefficients of median house prices, population density, and POI density in 2021, 2024, and the difference between 2021 and 2024.

4. Discussion: reflections on London's public EVCI rollout and EV adoption

The results of our spatial analyses reveal substantial geographic heterogeneity in both public Electric Vehicle Charging Infrastructure (EVCI) deployment and EV adoption across London. Our SAR results complement and contextualise the OLS findings.

At the local level, GWR's maps of R^2 (Figure 3) underscored that some areas - particularly western and southern London boroughs - are better explained by our chosen socio-economic, housing, and market sets of variables. For instance, wards in western London showed relatively high local R^2 for both public EVCI accessibility in 2024 and supply increase between 2021 and 2024, suggesting that the model's predictors account for a large share of the variation there. In contrast, eastern areas exhibited lower local R^2 , implying that other unobserved factors such as micro-political decisions, local constraints, or hyperlocal infrastructure may be at play.

The spatially nuanced results described in Section 3 carry significant implications for policy and planning. First, wards with low local R^2 or large areas of non-significance likely represent blind spots where neither socio-economic nor market variables explain EVCI rollout factors such as local grid capacity, small-scale politics, or conditional planning consents may be more important. Policymakers should target these neighbourhoods for further investigation or pilot interventions. Second - in

contrast to intuitive expectations - the positive correlation between house prices and charger deployment underscores an income-driven rollout only in some parts of the city (Figure 7): wealthier areas receive more infrastructure, while deprived or lower-ASG wards lag behind in western and southern London, but not in the central or eastern parts of the city. To prevent inequitable outcomes, local authorities might consider subsidies or minimum charging-station requirements in new rental developments or in areas identified as deprived with low accessibility gains, especially in central and west London (see Figure 5 and Figure 6). Third, analysing the change from 2021 to 2024 highlights where equity gaps are narrowing or widening. For example, also referring back to our research questions: “do current strategies for charging infrastructure sufficiently address equity?” and “And if not, where should policymakers focus interventions to ensure that socially and economically vulnerable communities are not left behind?” (Section 1), east London’s positive correlation between deprivation and accessibility improvement suggests that targeted interventions there are having an effect, whereas west London’s negative association may require renewed policy focus to ensure that deprived wards do not fall further behind (Figure 8).

Public EVCI Accessibility difference 2021-2024
Share of households deprived in 3+ dimensions

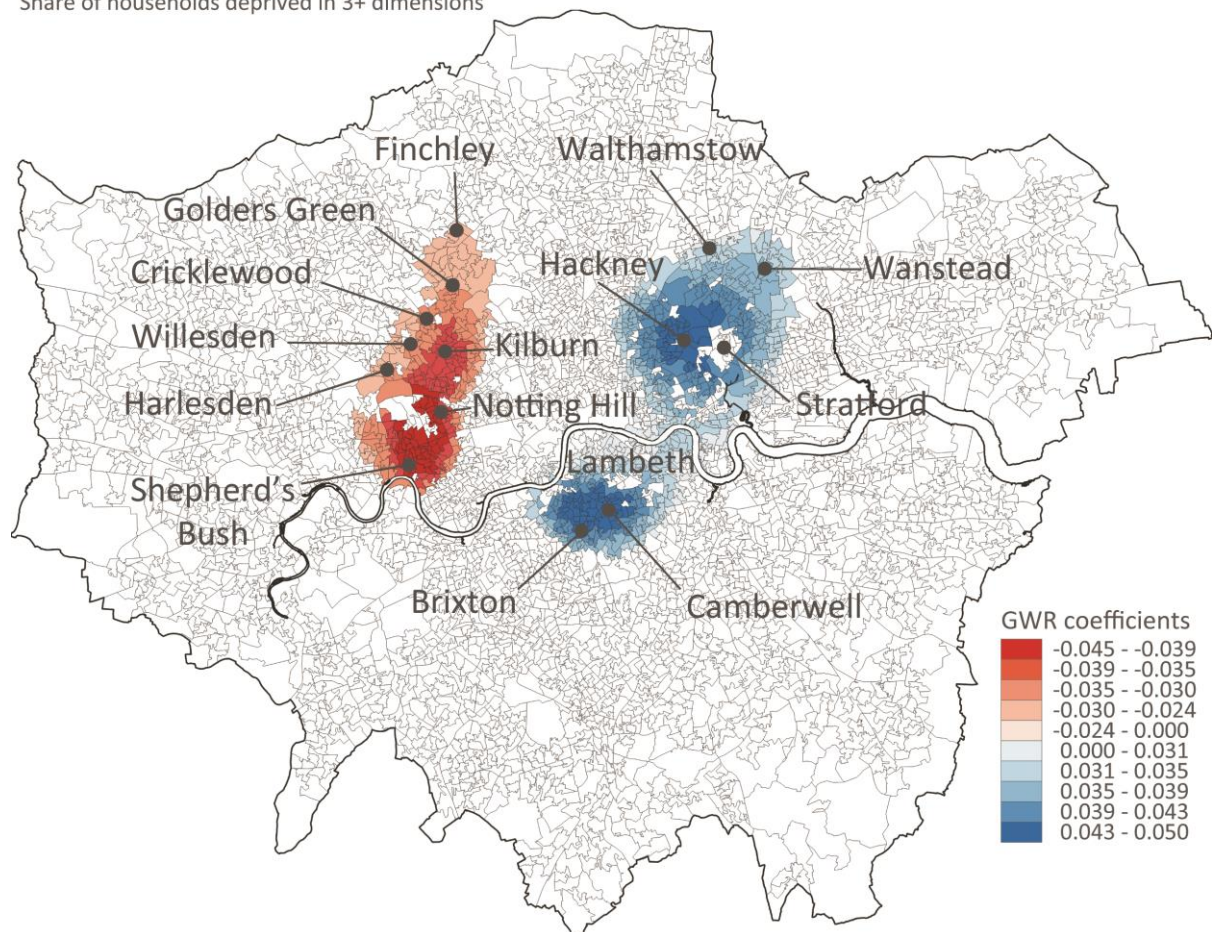


Figure 8 GWR coefficients of deprivation index for the difference in accessibility between 2021 and 2024.

Several unexpected or counterintuitive patterns emerged. Similarly to employment density and road network density - which were not included in the present study as (surprisingly) never statistically significant in any of the analyses, POI density - despite theoretical links to footfall and destination charging - is almost never a significant driver, suggesting that public chargers remain more residentially oriented in this early stage of infrastructure rollout, assuming that people mainly charge at home location, which may change in future scenarios when more EVCI are deployed to public

spaces like gyms, supermarkets etc. The negative correlation between population density and EV adoption in many wards indicates that infrastructure alone may not overcome barriers in cramped urban environments, such as a lack of charging access for on-street parkers or restrictive permit schemes. Housing tenure effects - where renters show enhanced access in some wards, but flats correlate negatively - underscore the need to address landlord inertia and the logistical complexities of multi-unit dwellings. These nuances highlight that a one-size-fits-all policy approach will likely fail to serve all communities equitably.

From an equity and policy perspective, a purely market-driven charging rollout risks perpetuating existing transport disadvantages (Roy and Law, 2022). Vulnerable communities - council housing residents, low-income households, and those without private parking - may be left behind without targeted interventions. Therefore, to answer the questions “do current strategies for charging infrastructure sufficiently address equity?” or “where should policymakers focus interventions to ensure that socially and economically vulnerable communities are not left behind?”, we recommend local authorities to perform spatially-explicit analyses to identify priority areas where to support policy enforcement of minimum charger requirements in rental and social housing, subsidies for deprived wards, and continuous monitoring to identify where equity gaps persist or widen over time – as the results of Section 3.3 demonstrated that it’s not possible to adopt one-size-fits-all kind of measures across different contexts in London.

Of course, there are a few relevant limitations to keep in mind. First, the adopted spatial resolution (LSOA) can mask finer-scale dynamics, especially in densely populated areas with mixed land uses and urban functions like the ones in inner London. Finer geographic units (e.g., output areas) or parcel-level analyses could reveal additional micro-spatial patterns, but data availability and confidentiality constraints often make that challenging. Second, the GWR and SAR models rely on accuracy in small-area measures - car ownership percentages, deprivation indices, and our accessibility scores. Measurement error in these variables could bias local coefficients and R^2 estimates. Third, the cross-sectional nature of our 2021 and 2024 snapshots captures associations, not causal effects; a positive correlation between house prices and charger deployment does not prove that raising house prices will bring chargers numbers up - it could be that a developer’s investment incentives drive both. Finally, by comparing only two time points, we may miss intermediate dynamics or lagged policy effects; but this is what is currently possible due to data availability.

These caveats suggest several paths for future research. A space-time framework - such as Temporal GWR (TGWR) (Fotheringham et al., 2015) - could model how local relationships evolve continuously, rather than only comparing two single years (provided that year by year data for each variable is available). Incorporating infrastructure variables like electrical grid capacity, substation locations, or zoning constraints may explain residuals in low- R^2 areas. At a finer scale, street-segment or parcel-level analyses could uncover how building typologies, parking configurations, or micro-demographic clusters influence both supply and adoption. On the other hand, the work could also be scaled up to regional or national scale, or transferred and applied to other cities to verify if the identified spatial patterns are generalisable or specific to the London metropolitan area. Finally, qualitative case studies in a few representative wards (especially those with low local R^2) could shed light on unobserved factors such as local politics, community engagement, or developer negotiations, that quantitative models (with the available data) cannot fully capture.

In summary, our combined OLS, SAR, and GWR analyses demonstrate that London’s EVCI rollout and EV adoption are governed by both global spatial processes and highly localised patterns. Recognising the interplay of market forces, socio-economic contexts, and housing characteristics is essential for designing equitable and effective policies. By targeting interventions in wards with persistent access gaps, and by continually monitoring how local coefficients shift over time, planners can ensure that the transition to electric mobility is both efficient and inclusive.

5. Conclusion

The present study set out to evaluate spatial equity in London's electric vehicle charging infrastructure by first confirming and modelling spatial dependence (via LM diagnostics and a Spatial Autoregressive model) and then uncovering local heterogeneity (through Geographically Weighted Regression). By combining these methods, we achieved both a global understanding of spillover effects - how charger deployment and EV registrations in one area influence neighbouring areas - and a fine-grained view of which socio-economic, housing, and market factors matter most in specific zones.

The significant LM and robust LM tests indicated that spatial dependence is non-negligible, justifying SAR estimation rather than ordinary least squares. By including a spatially lagged dependent variable, the SAR model confirms that an area's chargers deployment or EV registrations are influenced by neighbouring areas, capturing a global diffusion effect. The SAR framework captures a global "spillover" effect - i.e., the degree to which a change in one zone influences adjacent zones - while the subsequent Geographically Weighted Regression (GWR) highlights how the strength and direction of predictors vary locally. Consequently, SAR only provides a single, city-wide coefficient for each predictor - masking local variability. GWR, in contrast, reveals where and why those relationships strengthen, weaken, or reverse. For instance, SAR might show that overall, house prices positively predict charger growth, but GWR pinpoints that effect as stronger in southeast and northwest London, and weaker (or even negative) in isolated pockets. Thus, combining SAR's confirmation of spatial dependence with GWR's local detail offers a more complete understanding than either method alone. Together, these models demonstrate that neither public EVCI supply nor EV adoption can be adequately understood through a single, city-wide set of coefficients.

Our analysis revealed significant spatial autocorrelation in both supply and demand for public EVCI, justifying the SAR approach. Yet SAR's single, city-wide coefficients masked pronounced neighbourhood-level variation: wealthier areas (higher median house prices) consistently experienced more rapid charger roll-out only in some areas of the city, while deprived or lower social-grade areas saw slower accessibility gains with specific spatial patterns. Car ownership, housing tenure, and population density also exhibited spatially varying associations that sometimes contradicted intuitive expectations - such as weak or negative links between points of interest and charger deployment. Areas with low local R^2 values indicate blind spots where unobserved factors (e.g., local grid constraints or planning idiosyncrasies) likely drive outcomes.

Ultimately, ensuring an equitable transition to electric mobility demands both robust spatial analysis and proactive, targeted policy measures. Without deliberate efforts to address local disparities, the green transition risks reproducing or even exacerbating existing social and spatial inequalities rather than dismantling them.

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